10.18

a. Create two new variables. MOTHERCOLL is a dummy variable equaling one if MOTHER EDUC >12, zero otherwise. Similarly, FATHERCOLL equals one if FATHEREDUC >12 and zero otherwise. What percentage of parents have some college education in this sample?

ANS:

```
> # (a)
> dat$mothercoll <- ifelse(dat$mothereduc > 12, 1, 0)
> dat$fathercoll <- ifelse(dat$fathereduc > 12, 1, 0)
>
> cat("Sample size:", nrow(dat), "\n")
Sample size: 428
> cat("Mothers' college education percentage:", mean(dat$mothercoll) * 100, "\n")
Mothers' college education percentage: 12.14953
> cat("Fathers' college education percentage:", mean(dat$fathercoll) * 100, "\n")
Fathers' college education percentage: 11.68224
>
> table(dat$mothercoll)

  0   1
376  52
> table(dat$fathercoll)

  0   1
378  50
```

As you can see, mothers' college education is 12.14953%, and fathers' college education is 11.68224%, which is lower than the former.(Dummy variable with value 1 means college education, whereas 0 means without college education.)

b. Find the correlations between EDUC, MOTHERCOLL,and FATHERCOLL. Are the magnitudes of these correlations important? Can you make a logical argument why MOTHERCOLL and FATHERCOLL might be better instruments than MOTHEREDUC and FATHEREDUC?

ANS:

```
> cor(dat[, c("educ", "mothercoll", "fathercoll")])
                educ mothercoll fathercoll
educ       1.0000000  0.3594705  0.3984962
mothercoll 0.3594705  1.0000000  0.3545709
fathercoll 0.3984962  0.3545709  1.0000000
> cor(dat[, c("educ", "mothereduc", "fathereduc")])
                educ mothereduc fathereduc
educ       1.0000000  0.3870198  0.4154030
mothereduc 0.3870198  1.0000000  0.5540632
fathereduc 0.4154030  0.5540632  1.0000000
```

As you can see, COV(EDUC,MOTHERCOLL) = 0.3594705 and COV(EDUC, FATHERCOLL) = 0.3984962, which means these two variables are positively correlated with education. Moreover, the values of correlation are large enough in practice, therefore, the magnitudes of these correlations important. The reason why MOTHERCOLL and FATHERCOLL might be better instruments than MOTHEREDUC and FATHEREDUC is that parents with college degrees will pay much more attention to their children's education, and parents with higher levels of education doesn't represent that they have higher educational backgrounds.

c. Estimate the wage equation in Example 10.5 using MOTHERCOLL as the instrumental variable. What is the 95% interval estimate for the coefficient of EDUC?

ANS:

```
> iv_c <- ivreg(
+     log(wage) ~ educ + exper + I(exper^2) |
+         exper + I(exper^2) + mothercoll,
+     data = dat
+ )
>
> summary(iv_c)

Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
    I(exper^2) + mothercoll, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-3.08719 -0.32444  0.04147  0.36634  2.35621

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.1327561  0.4965325  -0.267  0.78932
educ         0.0760180  0.0394077   1.929  0.05440 .
exper        0.0433444  0.0134135   3.231  0.00133 **
I(exper^2)  -0.0008711  0.0004017  -2.169  0.03066 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147,      Adjusted R-squared: 0.1409
Wald test:   8.2 on 3 and 424 DF,  p-value: 2.569e-05

> confint(iv_c, "educ", level = 0.95)
            2.5 %    97.5 %
educ -0.001219763 0.1532557
```

The 95% confidence interval estimate for the coefficient of EDUC is

[-0.001219763 , 0.1532557].

d. For the problem in part (c), estimate the first-stage equation. What is the value of the F-test statistic for the hypothesis that MOTHERCOLL has no effect on EDUC? Is MOTHERCOLL a strong instrument?

ANS:

```
> # (d)
> fs_d <- lm(
+     educ ~ mothercoll + exper + I(exper^2),
+     data = dat
+ )
>
> summary(fs_d)

Call:
lm(formula = educ ~ mothercoll + exper + I(exper^2), data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-7.4267 -0.4826 -0.3731  1.0000  4.9353

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.079094   0.303118  39.849  < 2e-16 ***
mothercoll   2.517068   0.315713   7.973 1.46e-14 ***
exper        0.056230   0.042101   1.336    0.182
I(exper^2)  -0.001956   0.001256  -1.557    0.120
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared:  0.1347,	Adjusted R-squared:  0.1285
F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13

> linearHypothesis(fs_d, "mothercoll = 0")

Linear hypothesis test:
mothercoll = 0

Model 1: restricted model
Model 2: educ ~ mothercoll + exper + I(exper^2)

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    425 2219.2
2    424 1929.9  1    289.32 63.563 1.455e-14 ***
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
> t_mothercoll <- summary(fs_d)$coefficients["mothercoll", "t value"]
> cat("First-stage F in part (d):", t_mothercoll^2, "\n")
First-stage F in part (d): 63.56307
```

The value of the F-test statistic for the hypothesis is 63.56307, which is much larger than the critical value. As a result, we can reject the null hypothesis Ho: MOTHERCOLL has no effect on EDUC and infer that MOTHERCOLL is a strong instrument.

e. Estimate the wage equation in Example 10.5 using MOTHERCOLL and FATHERCOLL as the instrumental variables. What is the 95% interval estimate for the coefficient of EDUC? Is it narrower or wider than the one in part (c)?

ANS:

```
> # (e)
> iv_e <- ivreg(
+     log(wage) ~ educ + exper + I(exper^2) |
+         exper + I(exper^2) + mothercoll + fathercoll,
+     data = dat
+ )
>
> summary(iv_e)

Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
    I(exper^2) + mothercoll + fathercoll, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819  0.3922213  -0.712  0.47714
educ         0.0878477  0.0307808   2.854  0.00453 **
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,      Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06

> confint(iv_e, "educ", level = 0.95)
          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

The 95% confidence interval estimate for the coefficient of EDUC is [0.02751845,0.1481769], which is narrower than the one in part (c).

f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of MOTHERCOLL and FATHERCOLL. Do these instruments seem adequately strong?

```
> # (f)
> fs_f <- lm(
+     educ ~ mothercoll + fathercoll + exper + I(exper^2),
+     data = dat
+ )
>
> summary(fs_f)

Call:
lm(formula = educ ~ mothercoll + fathercoll + exper + I(exper^2),
    data = dat)

Residuals:
    Min      1Q  Median      3Q     Max
-7.2152 -0.3056 -0.2152  0.7627  5.0620

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.890259   0.290251  40.965  < 2e-16 ***
mothercoll   1.749947   0.322347   5.429 9.58e-08 ***
fathercoll   2.186612   0.329917   6.628 1.04e-10 ***
exper        0.049149   0.040133   1.225    0.221
I(exper^2)  -0.001449   0.001199  -1.209    0.227
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared:  0.2161,	Adjusted R-squared:  0.2086
F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16

>
> linearHypothesis(
+     fs_f,
+     c("mothercoll = 0", "fathercoll = 0")
+ )
```

```
Linear hypothesis test:
mothercoll = 0
fathercoll = 0

Model 1: restricted model
Model 2: educ ~ mothercoll + fathercoll + exper + I(exper^2)

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)    
1    425 2219.2                                  
2    423 1748.3  2    470.88 56.963 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

ANS:

The value of the F-test statistic for the hypothesis is 56.963, which is much larger than the critical value. As a result, we can reject the null hypothesis Ho: MOTHERCOLL and FATHERCOLL have no effect on EDUC and infer that MOTHERCOLL and FATHERCOLL are adequately strong.

f. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

ANS:

```
> # (g)
> summary(iv_e, diagnostics = TRUE)

Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper + 
    I(exper^2) + mothercoll + fathercoll, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-3.07797 -0.32128  0.03418  0.37648  2.36183 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)   
(Intercept) -0.2790819  0.3922213  -0.712  0.47714   
educ         0.0878477  0.0307808   2.854  0.00453 **
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 * 

Diagnostic tests:
                 df1 df2 statistic p-value    
Weak instruments   2 423    56.963  <2e-16 ***
Wu-Hausman         1 423     0.519   0.472    
Sargan             1  NA     0.238   0.626    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,      Adjusted R-squared: 0.147 
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06
```

The Sargan test in Diagnostic test above shows that the test ststistic NR^2 = 0.238 and the p-value = 0.626，which is bigger than the significance level = 0.05. Therefore, we cannot reject the null hypothesis Ho: All the instrument variables are exogeneous and are not correlated with the error term. It represents that our instrument variables are valid.